Any of these impacts can upset food webs, often dramatically, and human activity is responsible for an increasing number of them.

Staying resilient

Canadian ecologist Crawford Stanley Holling first proposed the idea of ecological resilience to describe the persistence of natural systems in the face of disruptive changes. Holling argued that natural systems require stability and resilience, but—contrary to what previous ecologists had assumed—these are not always the same qualities.

A stable system resists change in order to maintain the status quo, but resilience involves innovation and adaptation. Holling wrote that natural, undisturbed systems are likely to be continually in a transient state, with populations of some species increasing and others decreasing. However, these population changes are not as important as whether the whole system is being fundamentally altered. The resilience of a system can be described either by the time it takes to return to equilibrium after a big shock or by its capacity to absorb disturbance.

One example that Holling studied was the fisheries of the Great Lakes in North America. A large tonnage of sturgeon, herring, and other fish was harvested in the early decades of the 20th century, but overfishing dramatically reduced the catches. Despite subsequent controls on fishing, populations in the Great Lakes did not recover. Holling suggested that the intense fishing had progressively reduced the resilience of the ecosystem.

Holling argued that ecological resilience is not always positive. If a freshwater lake experiences a large input of nutrients from agricultural fertilizers, for example, it will become eutrophic: algae will thrive, depleting the lake’s oxygen and making it unsuitable for fish. Such a lake may be resilient, but it will become less biodiverse. Holling claimed that three critical factors determine resilience: the most a system can be changed